

EVELYN NAHYEON KIM

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1 EDUCATION

University of California, Los Angeles (UCLA) CA, US
Ph.D. in Computer Science, Advisor: Prof. Khalid Jawed and Prof. Wei Wang 2025 - Present

University of California, Los Angeles (UCLA) CA, US
M.S. in Mechanical Engineering, Advisor: Prof. Khalid Jawed 2023 - 2024

Seoul National University South Korea
B.S. in Mechanical Engineering and B.B.A. in Entrepreneurship 2018 - 2023

2 RESEARCH EXPERIENCE

Autonomous Driving for Sidewalk Delivery Robot CA, US
Ph.D. Student, Vision and Autonomy Intelligence Lab (VAIL) Jul 2026 – Present
Advisor: Prof. Bolei Zhou, UCLA

- ◇ Design a reward function for RL post-training of autonomous navigation models for Coco sidewalk delivery robots, incorporating action safety, stability, and social compliance.

Autonomous Driving Dataset Collection System in Large-Scale Farm CA, US
Ph.D. Student, Structures-Computer Interaction Lab Sep 2025 – Jun 2026
Advisor: Prof. Khalid Jawed and Dr. Jungseock Joo, UCLA

- ◇ Development of an autonomous mobile robotic system with multimodal sensing for large-scale agricultural data collection, mapping, and radio-based operation in remote farms with limited Internet connectivity.
- ◇ Long-term autonomous deployment in Fargo, ND and Oxnard, CA, repeatedly navigating consistent routes throughout crop growth cycles to collect large-scale multimodal datasets.

Adaptive Image Composition of two Diffusion Models South Korea
Research Engineer, CONNECTEVE Inc. Jul 2024 - Aug 2025
Advisor: Prof. Duhyun Ro, Seoul National University

- ◇ Develop a framework for composite image generation to address data scarcity in medical X-ray datasets.
- ◇ Introduce a novel framework that jointly denoises two DDPM models with a resampling mechanism to generate realistic and semantically consistent composite images with improved object-background coherence.

Self-Supervised Viewpoint Selector for Neural Radiance Fields (NeRF) CA, US
M.S. Student, Structures-Computer Interaction Lab Sep 2023 - Jun 2024
Advisor: Prof. Khalid Jawed, UCLA

- ◇ Improve NeRF training efficiency by introducing Self-supervised NeRF Image Selection (SNIS), a novel strategy for optimal camera viewpoint selection using a pseudo-label formulation analogous to reinforcement learning rewards.
- ◇ Integrate NeRF training with Unity-based camera control to establish a scalable self-supervised pipeline for viewpoint optimization and data collection.
- Master's Thesis, UCLA, 2024

Visual Positioning System (VPS) with Machine Learning South Korea
Research Engineer, VR Crew Inc. Jan 2023 - Aug 2023

- ◇ Develop a visual positioning system (VPS) for a VR application, estimating user location using mobile devices.
- ◇ Implement VR applications using a multimodal localization pipeline integrating dense LiDAR data, key localization modules, and global descriptor learning
- Patent, 2023

Knowledge Distillation with Network Inversion, Domain Adaptation South Korea
Research Intern, Korea Institute of Science and Technology (KIST) Aug 2021 - Oct 2022
Advisor: Prof. Suhyun Kim, KIST Data Science Team

- ◇ Propose a performance restoration method for pruned networks using knowledge distillation with synthetic data generated via network inversion.
- ◇ Investigate domain shrinkage effects during pruning and apply domain adaptation to improve sub-network performance.
- Bachelor's Thesis, Seoul National University, 2022

Optical Character Recognition (OCR)

Research Intern, Saige Research Inc.

South Korea
Feb 2021 - Aug 2021

Advisor: Prof. Frank Park, Seoul National University

- ◇ Develop and customize OCR models to automate manufacturing processes.
- ◇ Build bilingual (Korean–English) OCR pipelines, including dataset generation, document understanding, character detection, and recognition.

3 PUBLICATIONS & PATENT

- ◇ **Kim, E.**, Lien, T., Philip, J., Joo, J., Jawed, M. K. (2026). “AgriSC: A Benchmark for Semantic Consistency in Large-Scale Agricultural Environments.” Under review at IEEE RA-L
- ◇ **Kim, E.** (2024). “Advancing Neural Radiance Fields through Self-Supervised NeRF Image Selector (SNIS).” Master’s Thesis, University of California, Los Angeles.
- ◇ Jung, Y., Choi, S., **Kim, N.** (2023). “Apparatus and method for performing visual localization effectively.” Korean Patent No. 10-2023-0054544, filed April 26, 2023, issued December 18, 2023.
- ◇ **Kim, N.** (2022). “Data-Free Class-Specific Pruning.” Bachelor’s Thesis, Seoul National University.

4 ADDITIONAL TECHNICAL EXPERIENCE

3D Mouse: Spatial Information Inputs 2021

- ◇ Design and prototype a 3D input device using inertial and ultrasonic sensors for spatial interaction.
- ◇ Awarded 3rd place at the 10th Creative Engineering Design Fair (Seoul National University).

Software Engineer, Datacrunch Global Inc. 2023

- ◇ Develop a data-driven B2B platform for logistics analytics, tracking inventory flows and user behavior.
- ◇ Manage scalable APIs using Python (Django) with frontend integration (React) and database systems (MySQL).

Software Engineering Intern, PSX Inc. 2020

- ◇ Develop frontend and backend systems for a stock trading platform website and a hybrid mobile application using Python (Django) and React Native.
- ◇ Design a recommendation algorithm for an OTT media service to provide personalized content suggestions.

5 SKILLS

Programming Languages: Python, C/C++, SQL, React, Bash, R, NodeJS

Technologies and Frameworks: PyTorch, ROS2, Huggingface, Tensorflow, Docker, OpenCV, Pandas, Git